

Assembly Manual

Open-Source Cable-Actuated Snake Rescue Tool with Rotation-Assisted Body Winding

Version 1.0 | Prototype-stage hardware documentation | Dummy-snake validation only

Safety notice: This hardware is an early-stage prototype. Assemble and test the tool only with a dummy snake or flexible rubber tube during prototype validation. Live-animal use must be restricted to trained snake rescuers or authorized wildlife personnel under applicable ethical approval, local wildlife regulations, and expert supervision.

1. Purpose and scope

This manual describes the assembly procedure for a low-cost cable-actuated snake rescue tool. The tool uses a long aluminium shaft, bicycle-style brake lever, steel actuation cable, spring-return jaw linkage, compliant soft-grip jaw pads, and a manually rotatable working-end assembly. The intended prototype operation sequence is initial body grip, secure body capture, working-end rotation, body winding around the gripper/shaft region, controlled head-restraint positioning, and spring-assisted release.

2. Main components

Component group	Examples
Structural parts	Aluminium main shaft, angled aluminium handle tube, rear grip/end cap
Actuation parts	Bicycle brake lever/trigger, steel brake cable, Bowden sleeve, cable stopper/clamp, adjuster
Jaw and linkage	Movable jaw/grip component, fixed jaw support, stainless steel linkage strips, pivot bolts, washers, spacers
Return mechanism	Extension spring for jaw reopening
Contact interface	Soft rubber or TPU gripping pads
3D-printed parts	Cable guides, side-base supports, insert components, wedge supports, working-end supports
Fasteners and consumables	Bolts, nuts, washers, lock nuts, adhesive, finishing consumables

3. Tools required

- Drill and suitable drill bits for clearing holes
- Allen keys/spanners for bolts and lock nuts
- Cable cutter or pliers
- Screwdriver
- Vernier caliper for dimensional checks
- Sandpaper or file for deburring
- Adhesive suitable for fixing soft jaw pads
- Personal protective equipment during fabrication and testing

4. Assembly instructions

The snake rescue tool is assembled in seven stages: preparation of fabricated and printed parts, main shaft assembly, operator handle and trigger assembly, cable-routing assembly, working-end jaw and rotating mechanism assembly, final cable tensioning and alignment, and functional checking before validation.

4.1 Preparation of fabricated and printed parts

Inspect all 3D-printed parts for dimensional accuracy, layer defects, blocked holes, rough surfaces, and warping. Compare the printed jaw components, side-base supports, cable-passing supports, insert-rod parts, wedge supports, shaft collars, and rotating working-end supports with the corresponding CAD files.

Clear bolt holes, cable-passing holes, and rod-insertion holes using a suitable drill bit where required. Lightly sand contact surfaces to remove printing burrs and to ensure smooth fitting with the aluminium shaft, rods, fasteners, and cable guides.

Check the soft gripping pads for correct size, thickness, seating, and adhesion before fixing them to the jaw contact surfaces.

Cut the aluminium tube used as the main shaft to the required length according to the CAD model. Deburr both ends of the tube to prevent cable damage and operator injury.

Check linkage rods and metallic strips for straightness and correct hole alignment. Arrange all bolts, nuts, washers, spacers, springs, cable clamps, and brake-cable components according to the bill of materials before final assembly.

4.2 Main shaft assembly

Position the printed insert-rod and side-base components on the main aluminium shaft according to the assembly layout.

Align the supports along the shaft axis to avoid twisting of the cable path and to maintain correct alignment during jaw actuation and working-end rotation.

Insert bolts and washers through the printed supports and tighten gradually. Do not over-tighten, because excessive force can deform or crack printed polymer components.

Mount the cable-guide supports along the shaft at regular intervals. These guides keep the actuation cable close to the shaft and reduce lateral movement during trigger operation.

After tightening the cable guides, visually check the cable path from the handle end to the working end to confirm that the route is straight and free from sharp bends.

4.3 Operator handle and trigger assembly

Fix the angled handle support near the operator end of the shaft using the printed side-base and rod-insert components.

Select the handle angle so that the user can operate the tool with one hand while keeping the wrist in a natural position.

Mount the bicycle-style brake lever on the handle and secure it firmly using the clamp screw. Position the lever so that the operator can pull the trigger without changing hand position.

Insert the brake cable through the trigger lever and seat the cable end properly inside the lever slot to prevent slipping during repeated actuation.

Before connecting the cable to the working-end jaw mechanism, pull the lever several times to confirm smooth movement and complete return to its original position.

4.4 Cable-routing assembly

Route the steel actuation cable from the brake lever along the shaft through the printed cable-guide supports.

Use a Bowden sleeve or cable-passing support wherever the cable passes through curved or contact-prone regions to prevent rubbing against the shaft or printed parts.

Keep the cable slightly tensioned during routing to avoid slack loops.

At the working end, pass the cable through the jaw-side cable guide and connect it to the movable jaw linkage using a cable stopper or clamp.

Do not fully tighten the cable at this stage. Leave a small amount of adjustment for final tensioning after the jaw linkage and spring-return mechanism are installed.

4.5 Working-end jaw and rotating mechanism assembly

Attach the fixed jaw support to the front printed base. Connect the movable jaw component using pivot bolts, washers, and lock nuts.

Tighten the pivot bolts only enough to remove lateral play while still allowing free jaw movement. Avoid excessive tightening because it increases friction and reduces jaw response.

Connect the linkage arms between the cable attachment point and the movable jaw. Use washers or spacers at linkage joints to maintain clearance between moving parts.

After linkage installation, manually open and close the jaw several times to confirm smooth motion and to check that no printed part interferes with the linkage path.

Attach the extension spring between the jaw linkage and the fixed support point. Orient the spring so that it automatically reopens the jaw when the trigger is released.

Check the spring through the full jaw movement range to ensure that it does not collide with linkage arms, cable, or printed supports.

Fix the soft gripping pads to the inner jaw contact surfaces. These pads provide a compliant contact interface and increase friction during dummy-snake gripping tests.

Mount the manually rotatable working-end assembly coaxially with the shaft. Check that the jaw assembly can rotate about the shaft axis while maintaining jaw alignment and grip contact.

Rotate the working-end assembly manually through the expected working range. During rotation, inspect the cable, spring, linkage arms, and jaw pads to ensure that no twisting, locking, rubbing, or cable interference occurs. This check is essential because working-end rotation produces the body-winding action around the gripper/shaft region.

4.6 Final cable tensioning and alignment

After the shaft, handle, cable, jaw, spring, and rotating working-end assemblies are installed, tension the actuation cable at the jaw-side cable stopper.

Pull the trigger slowly while observing jaw closure. Adjust cable tension until the jaw closes fully under trigger operation and reopens automatically when the trigger is released.

Do not over-tighten the cable. Excessive preload can keep the jaw partially closed and reduce spring-return performance.

Check jaw opening, jaw closure, trigger return, spring return, and working-end rotation repeatedly.

Tighten all fasteners using hand tools. Use lock nuts at moving joints where possible to reduce loosening during repeated actuation.

Inspect the final assembly for sharp edges, exposed cable ends, loose fasteners, cable rubbing, excessive joint play, jaw pad detachment, and working-end rotation interference. Trim and cap exposed cable ends to prevent injury.

4.7 Functional check before validation

Operate the tool without load for repeated open-close cycles. Observe the trigger, cable, linkage, spring return, jaw pads, and manually rotatable working-end assembly during each cycle.

Test the jaw using a flexible rubber tube or dummy snake. Engage the dummy snake at a body region using the soft-grip jaw pads and actuate the brake-style trigger to achieve initial body grip and secure body capture.

After the dummy snake body is held, rotate the working-end assembly manually. This rotation should cause the captured dummy snake body to rotate and wind around the gripper/shaft region.

After rotation, release the trigger to confirm that the spring-return jaw opens and that the dummy snake can be removed without sticking.

The tool is ready for characterization only after the jaw opens and closes smoothly, the spring returns the jaw reliably, the cable remains aligned during actuation, the rotating working-end assembly moves without interference, the soft jaw pads remain fixed, and no structural looseness is observed during dummy-snake handling.

5. Pre-validation checklist

- All sharp edges removed and exposed cable ends capped
- Jaw opens and closes smoothly without binding
- Spring returns jaw reliably after trigger release
- Cable route is aligned and free from sharp bends
- Working-end assembly rotates without cable, spring, or linkage interference
- Soft jaw pads remain fixed during dummy-snake gripping
- Fasteners are tight and lock nuts are used at moving joints where possible
- Dummy snake can be gripped, rotated, body-wound, and released without sticking

6. Safety and limitation statement

This manual supports prototype assembly and dummy-snake functional checking only. The controlled head-restraint positioning and body-winding behavior described in this project have been demonstrated only using a flexible dummy snake/rubber tube. They should not be interpreted as validated live-snake restraint performance. Live-animal testing or deployment should only be performed by trained and authorized personnel under appropriate ethical approval, local wildlife regulations, and expert supervision.